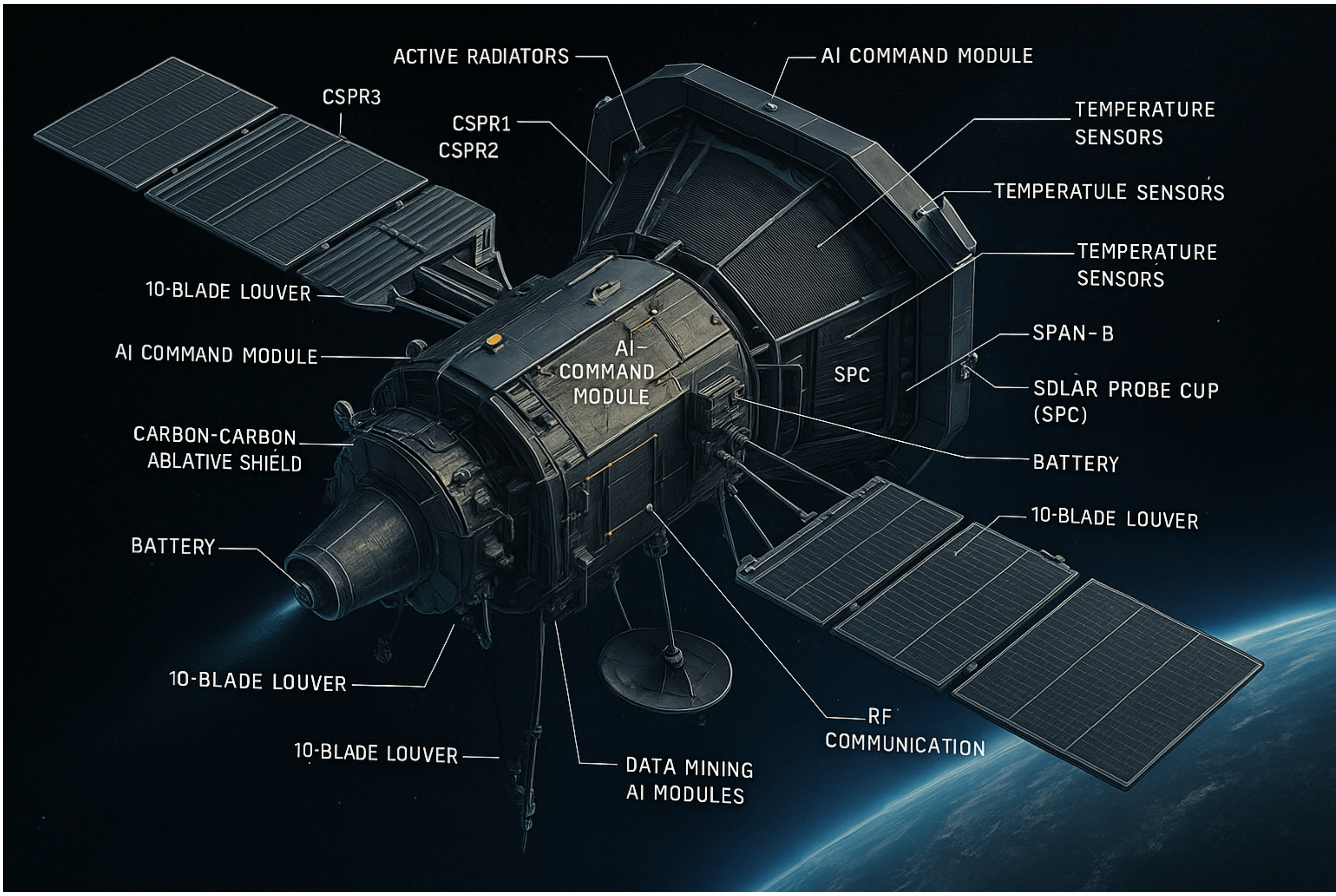
Combustion systems & fuel, composites, mechanical designs



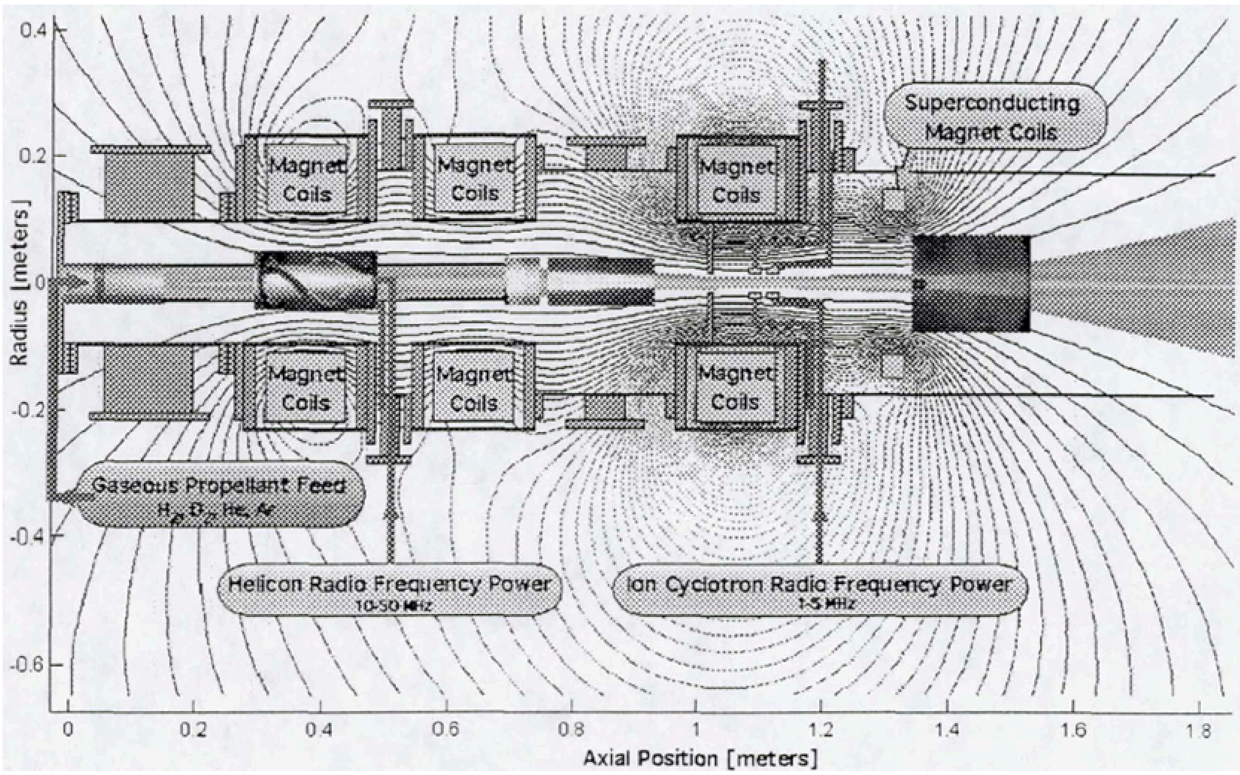
VASIMIR Propulsion

High density (>1019/m-3 ,) steady-state helicon discharges in hydrogen, helium, deuterium and argon are now routine and, as shown in Figures 3(a) and 3(b,) already show a high velocity exhaust downstream of the magnetic throat. Flow velocities of 15 Kml sec, measured with a reciprocating Mach probe, are quite common and indicate a respectably high specific impulse even before ICRH power is added.

JET ENGINE COMPONENTS

1. Diffuser - Slows down airflow and increases pressure.
2. Compressor - Increases air pressure via energy transfer from turbine.
3. Combustion Chamber - Location of air-fuel combustion.
4. Turbine - Converts thermal energy to mechanical energy.
5. Nozzle - Accelerates exhaust gases for thrust.

SPACECRAFT COMPONENTS
1. Jet Engines - Provide thrust.
2. Fuel System - Stores/supplies fuel.
3. Control System - Regulates engines/operations.

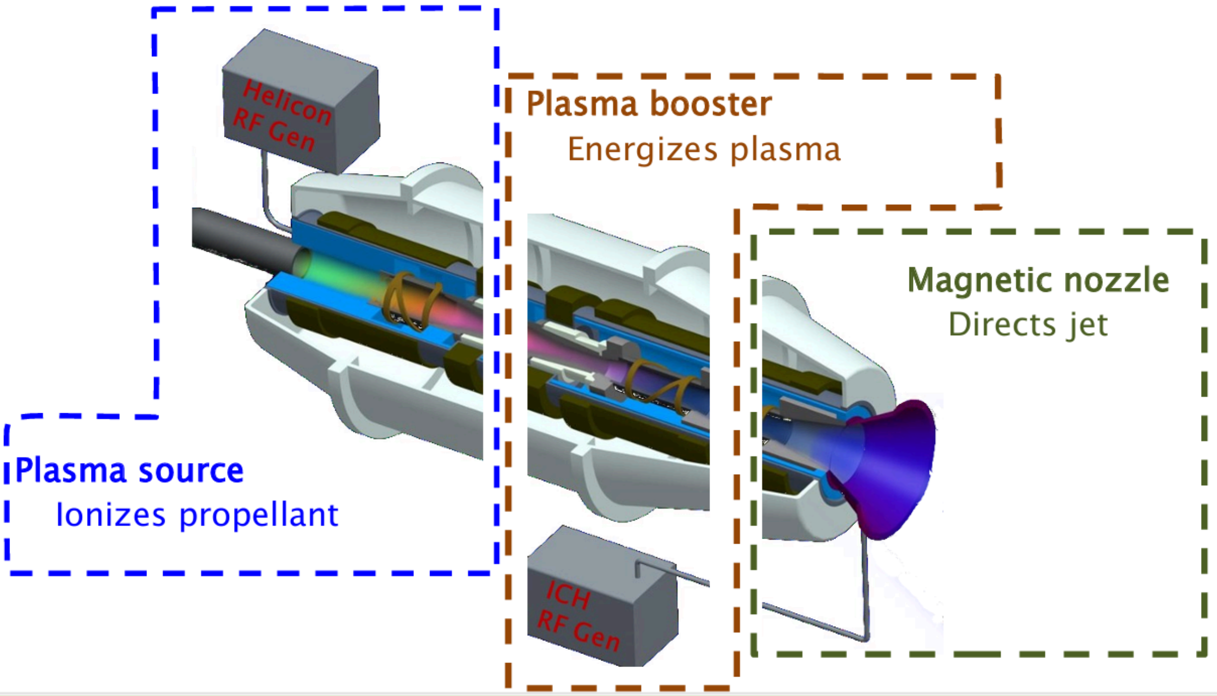


In the International Space Station, an experimental VASIMR operating at 25kW could provide sufficient thrust to neutralize the atmospheric drag on the facility. The system could operate on deuterium or hydrogen. VASIMR technology enables very fast human planetary transits to Mars and beyond. The development of the VASIMR and other advanced plasma rockets addresses a fundamental need in space transportation. Such an investment in technology will also benefit the commercial satellite industry, by enabling much higher payloads and mission flexibility. While a relative newcomer to the family of advanced rockets, V ASIMR has evolved rapidly and its research team is now demonstrating important i results in several areas. Such advances place it in a competitive realm with the more mature technologies of electrostatic plasma accelerators.

Receives power from solar or nuclear source: Directed plasma jet & generates thrust
Propellant options: H, He,
Ne, Ar, Kr, Xe…

Heated ions enter an adiabatic “nozzle” in the static magnetic field
◦ Directs ~90% of the ion energy into an

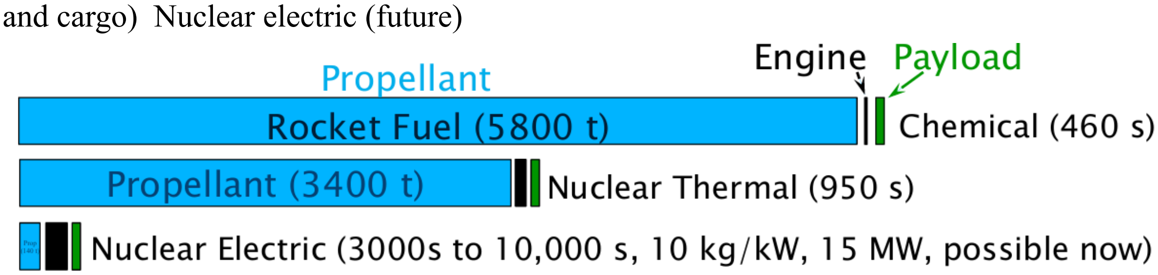
axial plasma jet for thrust

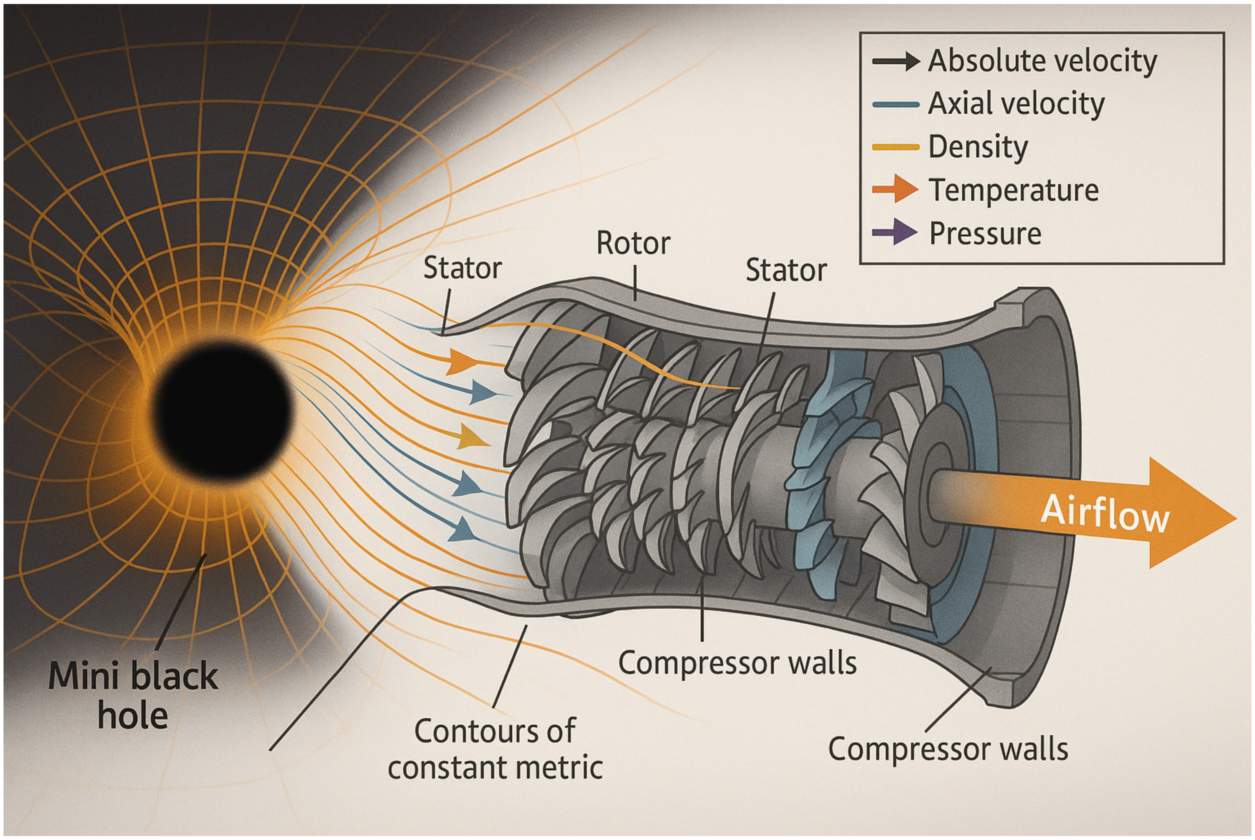


Uses RF waves to couple & power; VASIMR technology provides electric propulsion (EP) with the power density needed for sustainable human exploration

◦ The technology scales naturally to power levels for robotic and human exploration (> 40kW solar at first, scalable to multi-Megawatts)
◦ Specific impulse from 2000 s to 5000 s
◦ Total dry mass is approximately 2600 kg for 1 MW
◦ Adaptable to cover a broad range of mission Requirements.

Scalable to multi MW power levels High power density (6 MW/m2) Solar electric (near term and cargo) Nuclear electric (future)





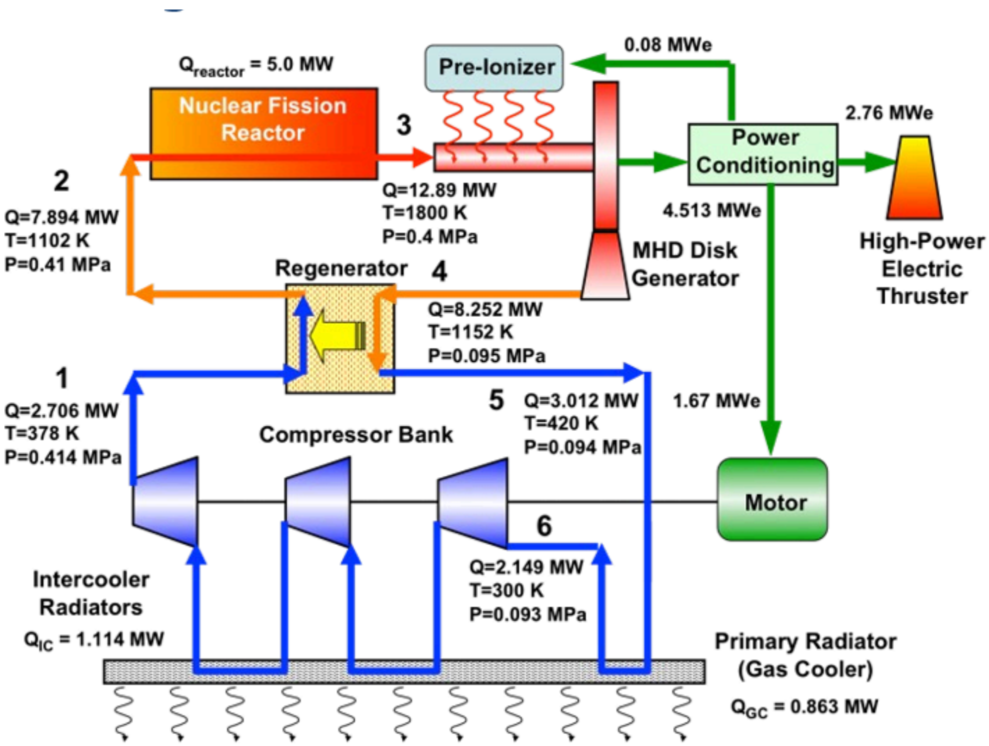
Radiator mass reduction for space drives the design to high temperatures (Stefan Boltzmann law) The High thermodynamic efficiency requires large ΔT Materials limit the maximum Temperature. The Surface density can approach 1 kg/m2 and specific mass below 0.2 kg/kW at 300 C

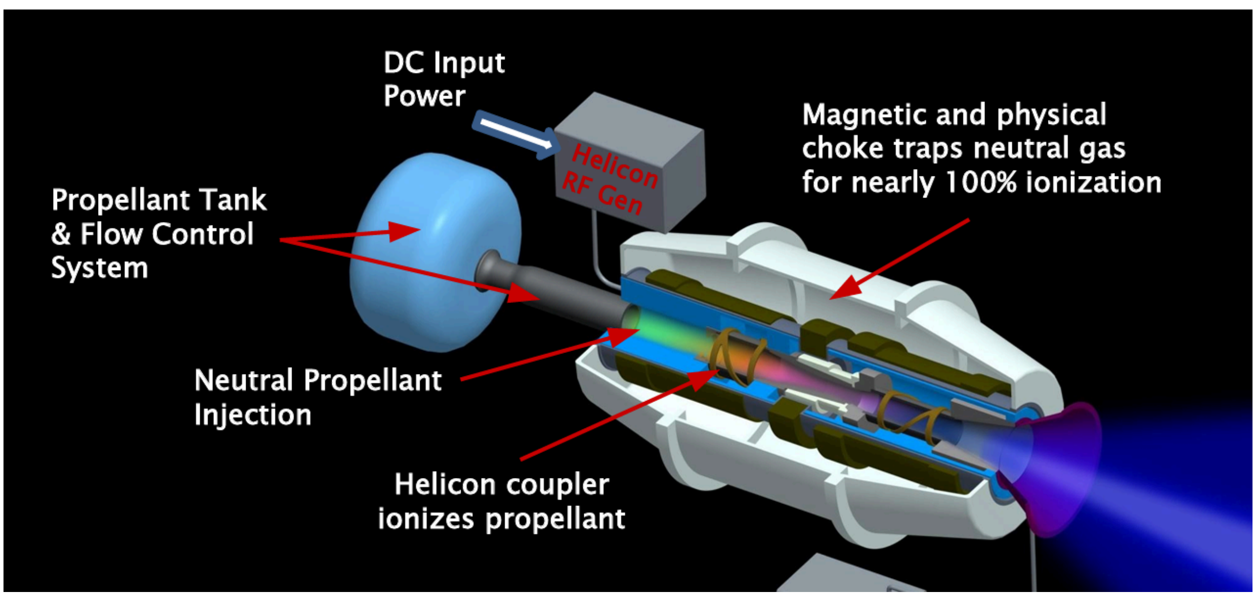
Steady-state RF PPU– Compact

Highly efficient ,Operated with both plasma stages TRL 5-6 (individual boards in vacuum)

Power rated: 48 kW
Efficiency: 91% (expect to increase to 95%)
Size: Weight:
40 cm x 40 cm x 120 cm
40.1 kg; α = 0.9 kg/kW
ICH

Power rated: 180 kW (24 hr ss burn in)
Efficiency:
98% now Size: Weight:
40 cm x 40 cm x 120 cm 87.1 kg; α = 0.5 kg/kW





Very high power density (~6 MW/m2) compared with DC discharges by coupling RF power through ceramic windows to excite naturally occurring plasma waves

No DC bias is applied, no cathodes or anodes are needed. Plasma facing materials are well insulated by strong magnetic fields with field strength of a few tesla

Specific impulse can be varied at constant power by adjusting the balance
between the plasma source and the ion cyclotron heating (ICH) booster sections

Wide range of operating power supported without loss of efficiency. Very efficient conversion of input power to jet power (75% thruster efficiency, >65% system efficiency) allows more thrust and less propellant usage than Hall thrusters for a given solar electric power plant

Wide range of propellant choices including xenon, krypton, argon, and neon. May also be adaptable to solids or liquids (ammonia, iodine, potassium, rubidium, etc.) if condensates do not pose a risk to the spacecraft.

Subsystem VASIMR® Hall
Gas Injection
- Single port injection
- Simple flow control
- Separate injection for anode and cathode
- Complex feedback required for flow control

Magnets
- Superconducting for high field with minimal power
- Well isolated from plasma
- Alignment with flow inherently provides surface protection
- Conventional, limits field
- Close proximity to plasma
- Alignment perpendicular to flow requires compromise for surface protection

Thermal Management
- Low-temperature fluid loop for power and avionics
- High-temperature fluid loop to manage thruster heat rejection
- Low-temperature fluid loop for power and avionics
- Very high-temperature to reject thruster waste heat by direct radiation

Called the “ion cyclotron” wave Ion Cyclotron Heating (ICH) uses resonance between wave and ion gyro-motion Resonance near the aft end of the
booster

Magnetic field fully shields ceramic parts
Acceleration from ICH enhances the source

PROPULSION MATERIALS
1. Fuel & Oxidizers - Generate thrust.
2. Plasma Propulsion - Plasma-based systems.

BLACK HOLE RESEARCH

1. Spacetime Symmetries
2. Plasma Tensor Converter - Analyzes plasma-black hole interactions.

THERMAL CONTROL SYSTEM (SACS)

- Heat dissipation: 6400 W at 9.86 Rs
- Liquid Cooling Loop (Redundant Pump/Electronics)
- Temp Ranges:
  - Operative: +20°C to +150°C
  - Survival: +10°C to +190°C (wet), -80°C to -130°C (dry)
- Activation:
  - Prelaunch: Accumulator heated to 40–50°C
  - Radiators 2 & 3 active at 0.89 AU (Day 41)

Key Components:

- Accumulator (non-redundant)
- Pump Electronics (block redundant)
- Platens, Radiators, Valves
- Thermal Simulation & Testing (C1/C2 configs)

CALCULATIONS: COMPRESSOR STAGES

Centrifugal Compressor:

INLET:

$C1 = 150$ m/s; $W1 = 250$ m/s; $U1 = 200$ m/s

OUTLET:

$C2 = 390$ m/s; $W2 = 150$ m/s; $U2 = 350$ m/s

$\beta 2$ = Flow angle; $\beta 2' = \beta 2 + 5°$

(a) Impeller Type:

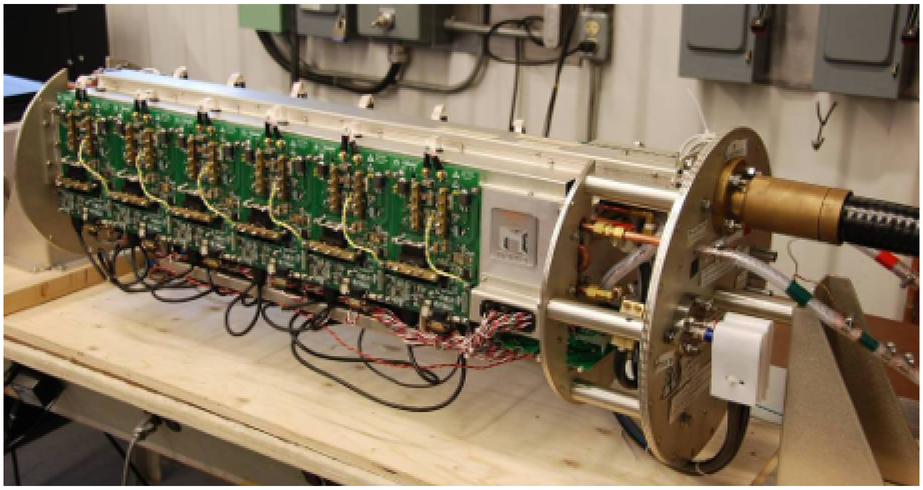
Based on the velocity triangle, calculate $\beta 2$.

If $\beta 2 < 90° \rightarrow$ Forward-curved

$\beta 2 = 90° \rightarrow$ Radial

$\beta 2 > 90° \rightarrow$ Backward-curved

Plasma propulsion systems (space & cars)



Notes: Experiments from 2009-2012 used a low
temperature superconducting magnet 6 K

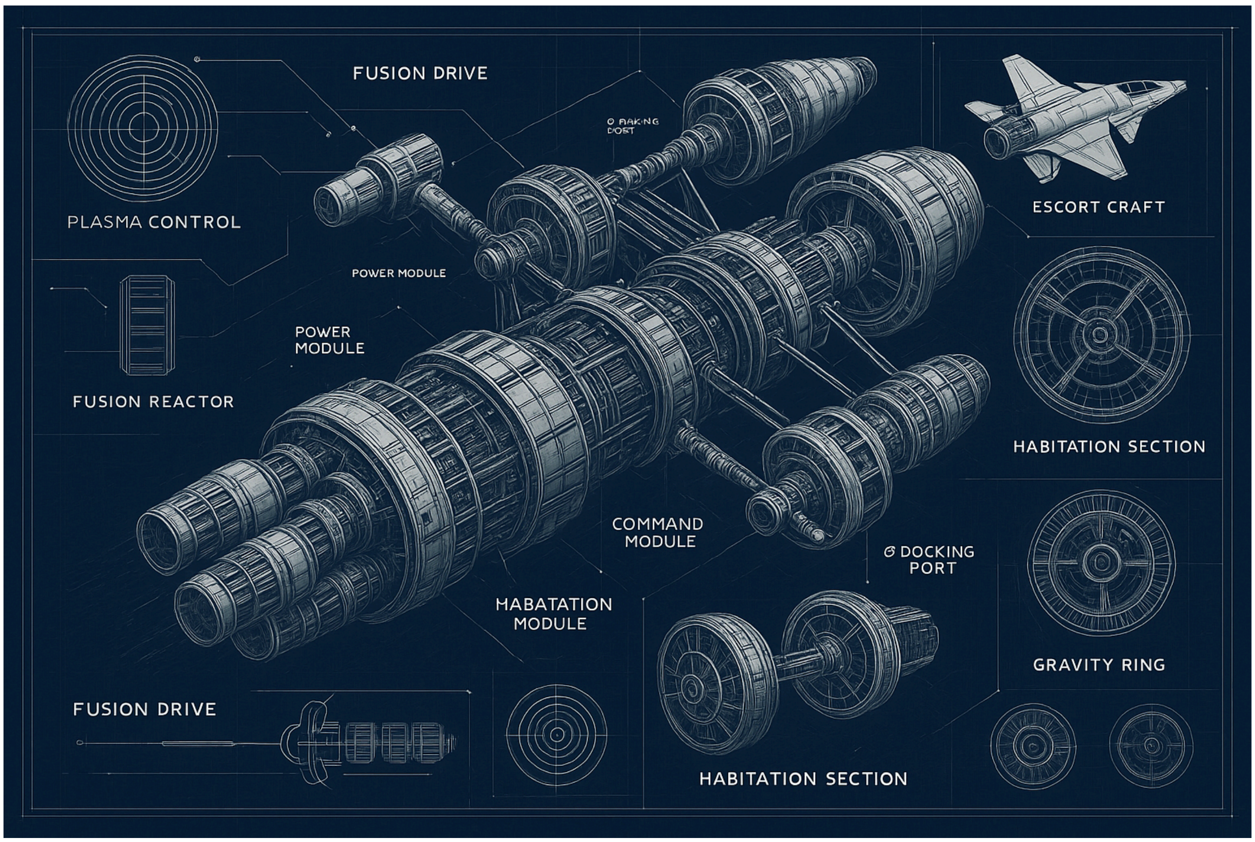
No coupling to plasma operation…Negligible current maintenance for steady magnetic field
High-temperature superconducting (HTS) magnet coil prototype developed and tested
in 2007 (Creare, Tai Yang) Preliminary design of proto-flight coils and
manufacturing test with 2nd generation HTS (SuperPower Inc.) in 2008 Performance improving with Moore's law Sunpower GT cryocooler baselined in 2013 preliminary design (200,000+ hours mean time to failure)

Notes: "Low temperature" system is pretty conventional ◦ Avionics, controllers, and cryocoolers with max temp ~35 °C ◦ RF generators with CSi modules can tolerate max temp up to ~85 °C to reduce radiator size "High temperature" system for rocket core ◦ Heat flux measured during VX-200 operation ◦ Rocket core has Integrated Cooling and Electrical (ICE) jackets ◦ Combination cooling function with the RF power couplers ◦ As high as 280 °C An infrared camera and thermocouples measured the temperature evolution inside the VX-200 plasma source during operation with no cooling. Transient analysis was used to identify heat flux and hot spots.

VASIMR® technology provides electric propulsion (EP) with the power density needed for
sustainable robotic space operations and human exploration [Reference 2,4]
◦ The technology scales naturally to power levels supported by the National Research Council for robotic and human exploration (> 40kW solar at first, scalable to multi-Megawatts)

VASIMR® technology differs from traditional Hall thrusters and ion engines in that it relies on
natural wave propagation in magnetized plasma to reach high power density.
◦ Waves include whistler or high-harmonic fast waves (HHFW) and ion cyclotron waves
◦ Relaxes Child-Langmuir and Debye shielding constraints encountered in DC discharges
The technology has a long history of development (1979-present) with collaboration of

academia (MIT, Princeton, UMD, UTexas, UMich, etc.), national laboratories (ORNL, LANL), international labs (Alfven, ANU, Japan), NASA and private industry.
◦ All of the physics to support VASIMR as an advanced propulsion technology are well understood and have
been demonstrated in the appropriate laboratory environment
◦ Experimental results have been presented and published in multiple peer reviewed journals over several years
◦ VASIMR® technology has been developed almost entirely with private funding (~$30M) and has been patented, but a great deal of misinformation circulates about its status
VASIMR is now fully enabled by modern technology developments, particularly high-power solid state transistors and high-temperature superconducting (i.e. YBCO) tape.



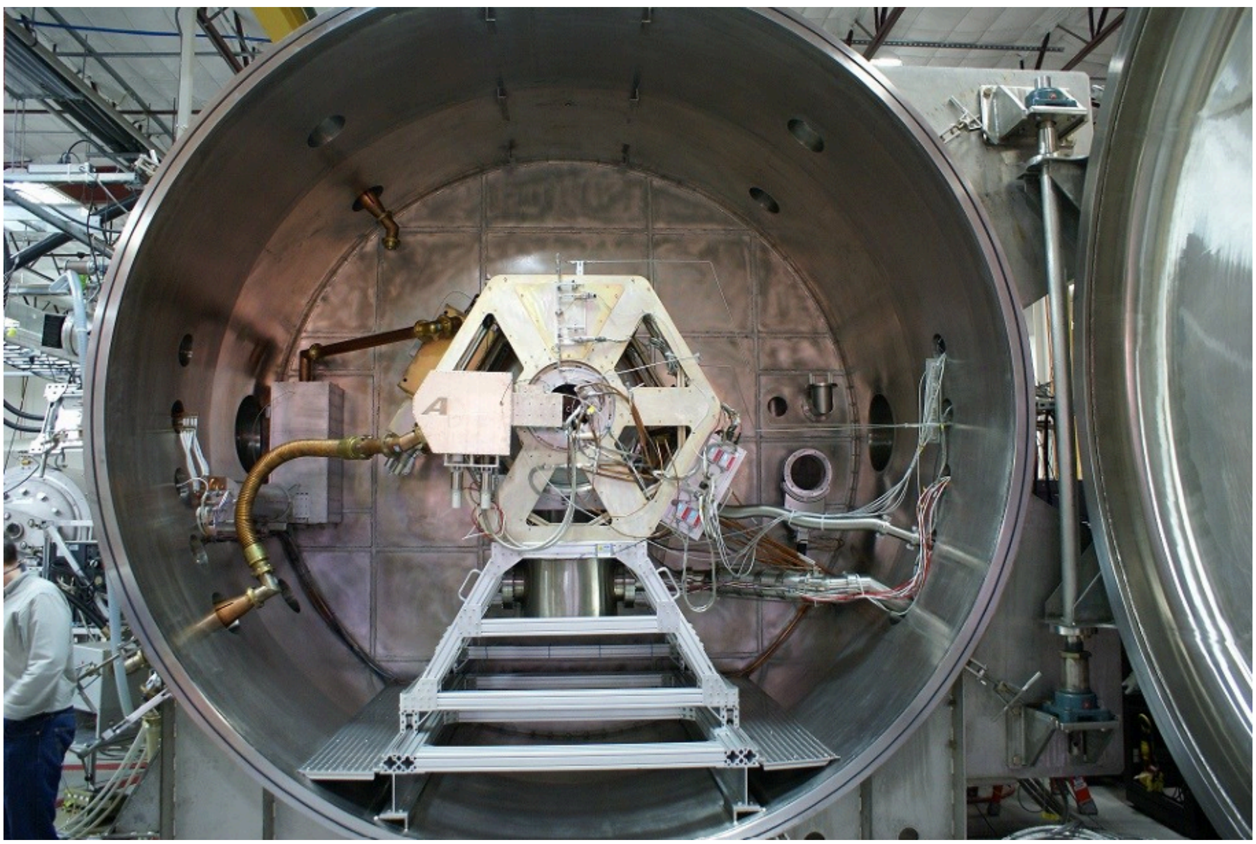
A. Ceramic Surface Temperature Measurements
A conceptual schematic of a VASIMR® system is shown in Fig. 2. The 1st stage “Ionizer” is a helicon type plasma source. The 2nd stage “RF Heater” is an ion cyclotron heating device. Figure 2 also shows the temperature measurement locations of plasma-facing ceramics in the VX-200SS™ prototype. The ceramic windows confine neutral gas and allow RF power to be coupled through to the plasma, but they must tolerate the associated heat load. Ceramic temperature measurements are therefore important for monitoring the

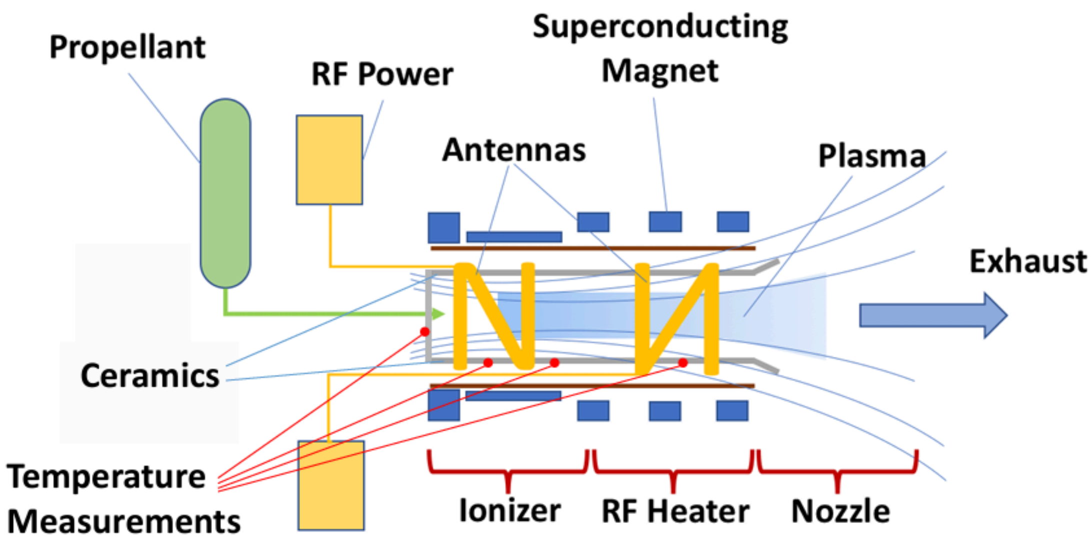
performance of the thermal design. Ad Astra has been using infrared pyrometers to measure plasma-facing component temperatures since 2019. Earlier efforts were thwarted by the challenges of the ambient environment within the rocket core.

B. Power Measurements

Each stage of a VASIMR thruster uses an independent radio frequency power system. Figure 3 shows a simplified schematic of the power circuit for the 2nd stage of the VX-200SS™ prototype. The circuit for The 1st stage is similar, except that the 1st stage power processing unit (PPU) is still outside the vacuum.

The locations where power can be measured in the circuit. In this paper, all presented power measurements were made with the DC current and voltage sensors located just to the left of the vacuum feedthrough. For the 1st stage circuit, the DC sensors are at the connection to the PPU.





Propellant Sealing The plasma-facing ceramics of the VX-200SS™ are an assembly of several manufactured components. The ceramic-to-ceramic joints in this assembly must be sufficiently sealed to prevent leakage of neutral gas into the space between the rocket core and the magnet cryostat. If the pressure in this annular space becomes too high, the operational power of the VX-200SS™ will be limited by Paschen breakdown at the RF antennas. Designing seals for this application has been challenging; the seals must tolerate the temperature cycles of the ceramics, electromagnetic fields in the proximity of the radio frequency antennas, and the static field of the superconducting magnet. Figure 6 presents two measurements of partial pressure versus time during VX-200SS operation: one made in May 2021 and another in June. These mass spectra were measured by a residual gas analyzer in the upstream section of the vacuum chamber. During thruster operation, the upstream argon (40 u) pressure is due to a combination of argon exhaust leakage through the vacuum chamber partition wall and neutral propellant leakage from the rocket core seals. Since the vacuum pumping and partition wall seals are typically unchanged from one experimental campaign to the next, the evolution of ambient upstream argon pressure during thruster operation is a good indicator of the rocket core seal quality.

Fusion reactor system: It was designed to analyze small and large aspect ratio, tokamak fusion
Inductive and non-inductive heating, driven
and ignited operation, burning DT, DD, or D3He fuels.

By pursuing peaked temperature and number density profiles fusion within the core of a plasma, a relatively small producing region was established, satisfying
Lawson and ignition criteria without necessitating large beta (13)throughout
the plasma. The lower temperature and number density outer regions would contribute to a
Volume-averaged 13 values within MHD stability limits.

Major radius (m)	2.48
Minor radius (m)	1.24
Aspect ratio	2.0
Elongation	3.0
Plasma volume (m^3)	225.8
Safety factor (edge)	2.50
Safety factor (axis)	2.08
Fuel ion density ($10^2/m^3$)	5.0
Electron density ($10^2/m^3$)	7.5
Plasma temperature (keV)	50
Volume averaged beta	0.318
Confinement and Plasma time (sec)	0.552
Average neutron wall load (MW/m^2)	1.03
Average radiation wall load (MW/m^2)	5.20
Ignition margin	1.235
Toroidal magnetic field (centerline) (T)	8.9
Maximum magnetic field (coil surface) (T)	52.1
Gain factor (Q)	73.1
Plasma current (MA)	66.22
Bootstrap current fraction	0.934
Wall reflectivity	0.98
Number density profile shape factor	1.0
Temperature profile shape factor	2.0